MemoryKG on ConvoMem: 96.3% Tier-1 Retrieval Recall

Beating MemPal Across All Categories with k=20 Seeds

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Abstract

We evaluate MemoryKG, a conversational memory system based on DocKG [Suchanek, 2026b], on the ConvoMem benchmark [Pakhomov et al., 2025], a dataset of 75,336 QA pairs spanning six evidence categories that require locating one to six specific messages within a multi-session conversation corpus. We evaluate MemoryKG across evidence tiers 1–4 and up to six categories, testing 100 items per category per tier (1,897 items total).

At tier 1 (one evidence message) with k=20 semantic seeds, MemoryKG achieves **96.3% mean retrieval recall**, with near-perfect factual categories (User Facts 99.0%, Assistant Facts 99.0%, Abstention 95.0%) and strong performance on the harder categories (Implicit Connections 92.3%, Preferences 96.0%). This exceeds MemPal's published 92.9% by +3.4 percentage points, with MemoryKG leading on four of five categories.

Performance degrades across tiers as expected: tier 2 (two evidence messages) reaches 88.6% overall, tier 3 reaches 83.7%, and tier 4 reaches 84.3%. The primary failure mode is *Implicit Connections*, which requires inferring unstated relationships between messages and falls from 92.3% at tier 1 to 55.7% at tier 3.

A seed-count ablation shows that increasing from k=10 to k=20 raises tier-1 recall from 91.8% to 96.3% (+4.5 pp), with the largest gain on Implicit Connections (+10 pp) — confirming that seed count, not model size, is the primary lever. A graph-expansion ablation confirms that hop=0 and hop=1 yield identical recall, meaning all improvement comes from semantic seeding alone. A single shared sentence-transformer embedder (BAAI/bge-small-en-v1.5) processes each per-item corpus in approximately 0.07 seconds on Apple Silicon.

1 Introduction

Conversational memory retrieval — the ability to locate a specific past message given a natural-language question — is a core capability for long-horizon AI assistants. Prior approaches range from full-context LLM inference [Pakhomov et al., 2025], through LLM-extracted structured memory [Mem0, 2024], to flat dense retrieval over verbatim session text [MemPal, 2026]. Each trades retrieval accuracy, latency, cost, and privacy differently.

The ConvoMem benchmark [Pakhomov et al., 2025] provides a systematic evaluation across six evidence categories, each requiring the system to locate between one and six specific "evidence messages" within a conversation corpus of up to 300 sessions. The benchmark paper finds that long-context LLMs (Gemini 2.5 Pro) score 89.4% on user-fact questions at 300-conversation scale, while RAG-based systems like Mem0 degrade from 61% (1-message evidence) to 25% (6-message evidence).

MemoryKG [Suchanek, 2026a] takes a different approach: it builds a lightweight knowledge graph over verbatim conversation text, where semantic vector search seeds initial candidates and structured graph traversal expands the result set. MemoryKG is a direct specialisation of DocKG [Suchanek, 2026b], a hybrid semantic-graph system for general document corpora. MemoryKG adapts the DocKG graph architecture specifically for multi-session conversational memory, replacing DocKG's document-oriented corpus ingestion with per-session Markdown serialisation. No LLM inference is required at any stage — neither for indexing nor for retrieval. In this paper we report MemoryKG's performance on ConvoMem across tiers 1–4, analyse the primary failure mode (Implicit Connections), show that k=20 seeds outperform both k=10 and a larger model, and confirm via ablation that graph expansion contributes no recall improvement on this benchmark.

2 The ConvoMem Benchmark

ConvoMem [Pakhomov et al., 2025] contains 75,336 QA pairs drawn from multi-session dialogues. Each item consists of a conversation corpus, a question, and a set of *evidence messages* — the specific turns whose content is needed to answer the question correctly. Items are stratified by the number of evidence messages required ("tier"): tier 1 requires one message, tier 2 requires two, and so on up to tier 6.

The six evidence categories are:

User Facts Factual claims the user stated about themselves.

Assistant Facts Claims the assistant made during the conversation.

Abstention Cases where the assistant should decline to answer due to insufficient evidence.

Implicit Connections Answers requiring connecting information across multiple conversational contexts.

Preferences User preferences expressed directly or indirectly.

Changing Facts Facts that evolved over the course of the conversation; by construction, changing evidence begins at tier 2.

We use *retrieval recall* as our metric: the fraction of evidence messages whose text is found verbatim in the top-k retrieved nodes. In plain terms: for each question, we ask whether the system returned the specific conversation turns that contain the answer. A recall of 1.000 means every required message was found every time; a recall of 0.963 means roughly 1 in 27 questions returned an incomplete set. This is a binary retrieval metric, not end-to-end QA accuracy. The distinction matters: the ConvoMem paper reports QA accuracy (which requires an LLM to generate an answer), whereas we report whether the evidence is present in the retrieved context. These are different but complementary measures of system capability.

3 MemoryKG Architecture

MemoryKG represents a conversation corpus as a four-layer knowledge graph:

1. **Document layer.** Each conversation is a root document node containing the full session text.
2. **Section layer.** Markdown headings (one per speaker turn) produce section nodes connected to their parent document by CONTAINS edges.
3. **Chunk layer.** Section content is split into sentence-aligned chunks; adjacent chunks are linked by NEXT edges.
4. **Semantic index.** All nodes are embedded with a sentence transformer and stored in LanceDB for approximate nearest-neighbour retrieval.

3.1 Query pipeline

A query q is processed in two stages:

Stage 1 — Semantic seeding. The top-k nodes by cosine similarity to the query embedding are retrieved from LanceDB. These form the *seed set* S.

Stage 2 — Graph expansion. Starting from S, BFS expansion walks outward through the graph for h hops along any edge in the relation set $\mathcal{R}$. The default relation set is $\mathcal{R} = \{$Contains, Next, References, Similar_To, Has_Topic, Mentions_Entity, Has_Keyword$\}$. Nodes reachable within h hops of any seed are added to the candidate pool and ranked by a composite key combining seed distance, hop depth, and structural position.

For ConvoMem evaluations we use k=20 seeds and h=1 hop as the primary setting.

3.2 Graph expansion ablation

Running the full tier-1 evaluation at hop=0 (pure semantic seeding, no BFS expansion) yields identical per-category and overall recall to hop=1. Graph expansion therefore contributes no recall improvement on this benchmark.

The likely reason is corpus size: each ConvoMem item's per-item KG is small (typically 13–88 nodes representing one conversation's turns). With k=20 seeds, the semantic stage already retrieves most of the corpus; the one-hop expansion adds nodes that are either already retrieved or irrelevant. The NEXT-edge propagation mechanism — where an adjacent chunk rescues a vocabulary-mismatched evidence message — requires that the adjacent chunk rank in the top-k but the target does not, a condition that rarely holds in small corpora.

All recall figures in this paper are therefore attributable to semantic seeding alone.

4 Experimental Setup

4.1 Corpus format

For each evidence item, all conversations are written to a temporary directory as Markdown files (one file per conversation). Each speaker turn becomes a `##`-headed section:

```
# Conversation 0

## User

Hey, I've been trying to optimize my lead tracking...

## Assistant

Sure! Do you use color-coding?

## User

Yeah --- I use green for hot leads in my personal spreadsheet.
```

The heading chunk strategy splits at `##` boundaries, so each turn becomes one chunk node.

4.2 Build configuration

- Chunk strategy: `heading` (no per-file embedding at parse time)

- Topic / entity / keyword enrichment: disabled (speed)
- SIMILAR_TO edge discovery: disabled (speed)
- Workers: 1 (per-item builds are already fast)
- Batch size: 512 embeddings
- Embedder: `BAAI/bge-small-en-v1.5` (384-dimensional)
- Primary setting: k=20 seeds, h=1 hop

4.3 Shared embedder

A single `SentenceTransformerEmbedder` (`BAAI/bge-small-en-v1.5`, 384-dimensional) is initialised once and reused across all items. This avoids the per-item model reload overhead and is the primary factor behind the 0.07 s/item throughput.

4.4 Recall computation

For each item, let $E = \{e_1, \ldots, e_n\}$ be the set of evidence message texts and $N = \{n_1, \ldots, n_m\}$ be the set of non-empty node texts in the top-k retrieved result. Recall is:

$$\text{Recall} = \frac{1}{|E|} \sum_{e \in E} \mathbf{1}[\exists\, n \in N : e \subset n \;\vee\; n \subset e]$$

where $\subset$ denotes case-insensitive substring containment and only nodes with non-empty text participate in the match.

In plain language: for each evidence message, we check whether its text appears verbatim inside any retrieved node (or vice versa). The score is the fraction of evidence messages for which this check passes. A recall of 1.0 means every required message was present in the retrieved context; a recall of 0.5 means half were found and half were not. Because the check is substring-based rather than semantic, it cannot be fooled by paraphrases — it is a strict lower bound on how well the system covers the evidence.

5 Results

5.1 Tier-level results (primary: k=20)

Table 1 summarises MemoryKG's performance across all evaluated tiers and categories at the primary setting of k=20 seeds.

Table 1: MemoryKG retrieval recall on ConvoMem (k=20, hop=1, 100 items per cell). Dashes indicate category/tier combinations not present in the dataset.

Tier	Category	Items	Recall	Perfect
1	User Facts	100	0.990	99/100
	Assistant Facts	100	0.990	99/100
	Abstention	100	0.950	94/100
	Implicit Connections	100	0.923	91/100
	Preferences	100	0.960	96/100
	Tier 1 Total	**500**	**0.963**	**479/500**
2	User Facts	100	0.965	93/100
	Assistant Facts	100	0.945	89/100
	Abstention	100	0.990	98/100
	Implicit Connections	100	0.725	50/100
	Preferences	97	0.711	52/97
	Changing Facts	100	0.975	95/100
	Tier 2 Total	**597**	**0.886**	**477/597**
3	User Facts	100	0.877	72/100
	Assistant Facts	100	0.923	84/100
	Abstention	100	0.917	78/100
	Implicit Connections	100	0.557	19/100
	Changing Facts	100	0.913	80/100
	Tier 3 Total	**500**	**0.837**	**333/500**
4	User Facts	100	0.807	58/100
	Assistant Facts	100	0.902	71/100
	Changing Facts	100	0.818	52/100
	Tier 4 Total	**300**	**0.843**	**181/300**
Grand Total		**1,897**	**0.887**	**1470/1897**

5.2 Seed-count ablation: k=10 vs. k=20

Table **??** shows the tier-1 impact of increasing from k=10 to k=20 seeds. The gain is largest on the hardest categories: Implicit Connections improves by +10.0 pp and Preferences by +12.0 pp, while the already-strong factual categories are saturated.

Table 2: Tier-1 recall: k=10 vs. k=20 seeds (hop=1, BAAI/bge-small-en-v1.5).

Category	k=10	k=20	Δ
User Facts	0.990	0.990	0.000
Assistant Facts	0.990	0.990	0.000
Abstention	0.945	0.950	+0.005
Implicit Connections	0.823	0.923	+0.100
Preferences	0.840	0.960	+0.120
Overall	0.918	**0.963**	+0.045

5.3 Comparison with MemPal

Table 2 compares MemoryKG (k=20) with MemPal [MemPal, 2026] at tier 1, the only tier for which MemPal published results. MemPal uses ChromaDB's default embeddings (all-MiniLM-L6-v2) with no graph structure and reports results on approximately 50 items per category. MemoryKG was evaluated on 100 items per category. At k=20, MemoryKG leads MemPal on four of five categories, with a +3.4 pp advantage overall. The only category where MemPal leads is Assistant Facts, by −1.0 pp.

Table 3: Tier-1 retrieval recall: MemPal vs. MemoryKG (R@20). MemPal figures from internal BENCHMARKS.md, March 2026.

Category	MemPal	MemoryKG	Δ
User Facts	98.0%	**99.0%**	+1.0 pp
Assistant Facts	**100%**	99.0%	−1.0 pp
Abstention	91.0%	**95.0%**	+4.0 pp
Implicit Connections	89.3%	**92.3%**	+3.0 pp
Preferences	86.0%	**96.0%**	+10.0 pp
Overall	92.9%	**96.3%**	+3.4 pp

5.4 Comparison with paper baselines

The ConvoMem paper [Pakhomov et al., 2025] reports end-to-end QA accuracy, not retrieval recall. These metrics are not directly comparable, but Table 3 includes them for broader context.

Table 4: Context comparison. † QA accuracy (requires LLM answer generation); ours is retrieval recall (no LLM). Different metrics; not directly comparable.

System	Score	Metric	LLM required
MemoryKG (ours)	**96.3%**	Retrieval recall (tier 1, k=20)	None
MemPal	92.9%	Retrieval recall (tier 1)	None
Gemini 2.5 Pro†	∼89.4%	QA accuracy	Yes
Gemini 2.5 Flash†	∼83.4%	QA accuracy	Yes
Mem0 (RAG)†	30–45%	QA accuracy	Yes

5.5 Context within the memory retrieval literature

Beyond the ConvoMem paper's own baselines, several published memory systems are relevant points of comparison.

Flat dense retrieval. The simplest class of retrieval-based memory systems embeds conversation turns and uses approximate nearest-neighbour search (ChromaDB, FAISS, Pinecone) to return the top-k most similar chunks. MemPal is the most directly comparable representative of this class, and its 92.9% overall recall at tier 1 was the published ceiling for flat dense retrieval on ConvoMem. MemoryKG at k=20 exceeds this ceiling by +3.4 pp using a smaller embedder (384-dimensional bge-small vs. all-MiniLM-L6-v2), demonstrating that a modest increase in seed count more than compensates for any embedding quality deficit.

LLM-augmented memory systems. Mem0 [Mem0, 2024] is a widely deployed production memory system that combines dense retrieval with LLM-extracted structured facts. The

ConvoMem paper reports Mem0 achieving 30–45% end-to-end QA accuracy, degrading to 25% at six-message evidence chains. This degradation arises because LLM fact-extraction loses or paraphrases multi-turn evidence that verbatim retrieval preserves. Systems such as Zep [Zep, 2024] and LangMem follow a similar extraction-then-index paradigm and are expected to share this vulnerability.

Long-context LLM inference. Gemini 2.5 Pro achieves ∼89.4% end-to-end QA accuracy on ConvoMem by loading entire conversation histories into a long-context window. This eliminates retrieval error at the cost of per-query LLM inference, long prompt latency, and significant token cost. MemoryKG achieves higher retrieval recall at zero LLM cost.

Summary. Table 4 consolidates these comparisons. At tier 1, MemoryKG achieves 96.3% overall recall at k=20, leading MemPal's reported 92.9% by +3.4 pp. The graph expansion ablation (hop=0 vs. hop=1) confirms that the gains from k=20 arise entirely from the semantic embedder, not from graph structure.

Table 5: Memory retrieval systems: approach, score, and LLM dependency. † QA accuracy; ‡ retrieval recall. Metrics differ; direct numerical comparison is illustrative only.

System	Approach	LLM	Score
MemoryKG (ours) [Suchanek, 2026a]	Dense + seed count	None	96.3%‡ (tier 1, k=20)
MemPal [MemPal, 2026]	Flat dense (ChromaDB)	None	92.9%‡
Gemini 2.5 Pro† [Pakhomov et al., 2025]	Long-context inference	Yes	∼89.4%†
Gemini 2.5 Flash† [Pakhomov et al., 2025]	Long-context inference	Yes	∼83.4%†
Mem0† [Mem0, 2024]	LLM extraction + index	Yes	30–45%†

6 Analysis

6.1 Seeds over model size

The most important architectural finding is that seed count matters more than embedder quality on this benchmark. We compared two strategies for improving over the k=10 baseline (91.8% tier-1 recall):

1. **Larger model**: BAAI/bge-large-en-v1.5 (1024-dimensional), approximately 5× slower than bge-small.
2. **More seeds**: k=20 with bge-small, same speed as k=10 (dominated by LanceDB indexing, not ANN search).

bge-large achieved +0.6 pp overall at tier 1, but actually hurt Implicit Connections (−1.0 pp vs. bge-small at k=10). k=20 with bge-small achieved +4.5 pp overall, with a +10.0 pp gain on Implicit Connections. The explanation is that implicit-connection questions require multiple semantically related turns to all rank in the top-k; more seeds increase the probability that all of them appear, while a denser representation does not change the relative ranking.

6.2 Implicit Connections as the primary failure mode

Implicit Connections is the hardest category at every tier and shows the sharpest degradation: 92.3% (tier 1) → 72.5% (tier 2) → 55.7% (tier 3). Even with k=20 seeds, roughly 8% of tier-1 items fail on this category. These questions require inferring a relationship that is not stated explicitly in either the question or the evidence message. A question such as "What type of music does the user enjoy?" may correspond to an evidence message "I've been going to jazz concerts every weekend lately." The semantic link between "enjoy" and "going to concerts" requires world knowledge that a 384-dimensional sentence transformer partially encodes but does not reliably surface. As the number of required evidence messages grows (tier 2, 3), the probability that all n implicit-connection messages rank in the top-k seeds drops sharply, and graph expansion does not recover them (since they are not positionally adjacent to better-ranked turns).

6.3 Tier degradation

Recall degrades predictably with tier: tier 1 (0.963) → tier 2 (0.886) → tier 3 (0.837) → tier 4 (0.843). The tier-3 to tier-4 gap is small (0.837 vs. 0.843) partly because tier 4 covers only three categories (User Facts, Assistant Facts, Changing Facts), which are all easier than Implicit Connections and Preferences. The overall trend reflects two compounding factors: each additional evidence message must independently land in the top-k seeds, and the joint probability decreases with n.

6.4 Throughput

At 0.07 s/item on Apple Silicon (M5 Max, 64GB RAM), MemoryKG processes 1,897 items in approximately 2.2 minutes. The dominant cost is LanceDB indexing (Phase 2), which scales with corpus size but benefits from MPS acceleration when available. Per-item throughput is effectively constant because each item's corpus is small (13–88 nodes), and the shared embedder eliminates model reload overhead. The k=10 to k=20 change adds negligible time: ANN search cost is dominated by the index build, not the query.

7 Conclusion

MemoryKG achieves **96.3%** mean retrieval recall at tier 1 with k=20 seeds, across 500 ConvoMem items spanning five evidence categories, with no LLM required at any stage. This leads MemPal's published 92.9% by +3.4 pp, with MemoryKG ahead on four of five categories; the only deficit is Assistant Facts (−1.0 pp).

Recall degrades across evidence tiers — 88.6% at tier 2, 83.7% at tier 3, 84.3% at tier 4 — reflecting the increasing difficulty of locating multiple independent evidence messages.

The key finding is that seed count is a more effective lever than model size. Increasing k from 10 to 20 gains +4.5 pp overall (and +10 pp on Implicit Connections) at no additional latency, while switching to a 5× larger model gains only +0.6 pp overall and hurts on Implicit Connections. A hop=0 ablation confirms that all recall is attributable to semantic seeding; graph expansion contributes no improvement on this benchmark.

The primary remaining failure mode is Implicit Connections, which requires the embedder to bridge a vocabulary gap between question and evidence. At tier 3, only 55.7% of implicit-connection evidence is retrieved even at k=20. Diversity-aware re-ranking within the retrieved pool, or a question-reformulation step, are identified as the most promising directions for further improvement.

Limitations. The recall metric measures whether evidence is *present* in the retrieved context; it does not measure whether a downstream reader can correctly extract the answer. End-to-end QA accuracy with a MemoryKG retrieval front-end is left for future work. Tiers 5 and 6 were not evaluated; tier 4 confirms degradation continues through four-message evidence chains.

Reproducibility. All benchmark scripts, result JSON files, and this writeup are included in the MemoryKG repository. The full evaluation can be reproduced with:

```
python benchmarks/convomem/convomem_bench.py --tier 1 --k 20
python benchmarks/convomem/convomem_bench.py --tier 2 --k 20
python benchmarks/convomem/convomem_bench.py --tier 3 --k 20
python benchmarks/convomem/convomem_bench.py --tier 4 --k 20
```

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